

MM INSTITUTE OF MANAGEMENT

Master of Business Administration (MBA) — 1st Semester

Subject: Data Analysis for Managers (MBA-103)

Assignment: Assignment - 1

Date of Assign: 09/09/2026

Date of Submission: 14/09/2026

Course Outcome: CO1 (Understand statistical tools & techniques)

Total Marks: 30 Marks (Sec A: 8, Sec B: 22)

SECTION A (Short Answer Questions — Compulsory, 8 Marks)

Q.1 (a) Explain the role of Statistics in decision-making.

[CO1 | PEO2 | PO3 — 2 Marks]

Statistics provides quantitative methodologies to convert raw data into actionable intelligence, serving as an indispensable foundation for managerial decision-making under risk and uncertainty. Its core roles include:

- **Evidence-Based Decision Making:** Eliminates guesswork and intuition by enabling executives to base strategies on verifiable, empirical evidence.
- **Market Forecasting & Planning:** Techniques such as regression and time-series analysis allow organizations to project customer demand, revenues, and industry trends.
- **Quality Control & Process Optimization:** Statistical Process Control (SPC) and Six Sigma methodologies utilize control charts and sampling to minimize operational variances and defects.
- **Risk Assessment & Mitigation:** Probability distributions model financial, market, and credit risks, enabling managers to construct optimal portfolios and scenario models.
- **Resource Allocation:** Quantitative analysis ensures efficient allocation of human capital, marketing budgets, and operational capacities to maximize return on investment (ROI).

Q.1 (b) Write the formula for Arithmetic Mean for a continuous frequency distribution.

[CO1 | PEO1 | PO3 | PSO1 — 2 Marks]

For a continuous frequency distribution, data is grouped into class intervals. The mid-value (m_i) of each class interval is computed as $m_i = (L_i + U_i) / 2$, where L_i is the lower class limit and U_i is the upper class limit.

1. Direct Method:

$$\bar{X} = (\sum f_i \cdot m_i) / N$$

2. Step-Deviation (Short-cut) Method:

$$\bar{X} = A + [(\sum f_i \cdot d'_i) / N] \times h$$

Where: $\bar{X}$ = Arithmetic Mean, A = Assumed Mean, f_i = Frequency of the i -th class, m_i = Mid-point of the i -th class, h = Class width (interval length), $d'_i = (m_i - A) / h$, and $N = \sum f_i$ (Total Frequency).

Q.1 (c) Explain the relationship between Mean, Median, and Mode.

[CO1 | PEO1 | PO4 | PSO1 — 2 Marks]

The relationship among the three measures of central tendency depends directly on the degree of skewness (asymmetry) of the frequency distribution:

- **Symmetrical (Normal) Distribution:** The distribution is perfectly bell-shaped. Mean, median, and mode coincide at the exact center:

$$\text{Mean} = \text{Median} = \text{Mode}$$

- **Positively Skewed Distribution (Skewed to the Right):** The tail pulls toward higher values, inflating the mean more than the median:

$$\text{Mean} > \text{Median} > \text{Mode}$$

- **Negatively Skewed Distribution (Skewed to the Left):** Extreme low values drag the mean down:

$$\text{Mean} < \text{Median} < \text{Mode}$$

- **Karl Pearson's Empirical Relationship:** For moderately skewed unimodal distributions, Karl Pearson established the widely used empirical approximation:

$$\text{Mode} = 3 \cdot \text{Median} - 2 \cdot \text{Mean} \quad \text{or} \quad \text{Mean} - \text{Mode} = 3 \cdot (\text{Mean} - \text{Median})$$

Q.1 (d) Define Range.

[CO1 | PEO1 | PO4 — 2 Marks]

Range is the simplest absolute measure of dispersion. It represents the numerical difference between the highest (maximum) value and the lowest (minimum) value in an observed data set.

$$\text{Range (R)} = L - S$$

Where **L** = Largest (maximum) observation, and **S** = Smallest (minimum) observation.

To compare variations across different distributions with unequal units or scales, the relative measure known as the **Coefficient of Range** is applied:

$$\text{Coefficient of Range} = (L - S) / (L + S)$$

Managerial Significance: Range provides immediate insight into overall variability, widely used in statistical quality control charts (e.g., X-bar and R-charts) and stock price volatility tracking.

SECTION B (Descriptive & Numerical Problems — 22 Marks)**Q.2 Define Statistics and explain its meaning, nature, and characteristics in detail.**

[CO1 | PEO1 | PO3 | PSO2 — 5.5 Marks]

1. Definition of Statistics

Statistics has been defined both in the plural sense (as numerical data) and in the singular sense (as statistical methods):

Definition by Horace Secrist (Plural Sense):

"By statistics we mean aggregates of facts affected to a marked extent by multiplicity of causes, numerically expressed, enumerated or estimated according to reasonable standards of accuracy, collected in a systematic manner for a predetermined purpose, and placed in relation to each other."

Definition by Croxton and Cowden (Singular Sense):

"Statistics may be defined as the science of collection, presentation, analysis, and interpretation of numerical data."

2. Nature of Statistics

- **Statistics as a Science:** It embodies a systematized body of scientific knowledge, utilizing standardized mathematical principles, formulas, laws of probability, and systematic steps of inquiry.
- **Statistics as an Art:** It requires professional judgment, practical skill, domain expertise, and analytical acumen to handle real-world ambiguities, interpret findings, and translate numbers into strategic managerial action.

3. Essential Characteristics of Statistics

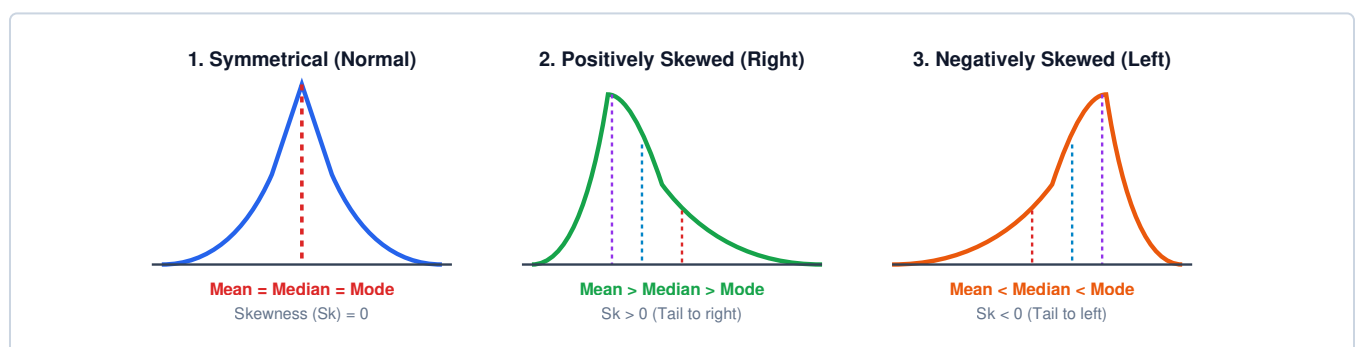
1. **Aggregate of Facts:** A single, isolated numerical figure (e.g., "Product A sells at \$50") is not statistics. Only a collection or series of related figures can be compared and analyzed.
2. **Affected by Multiplicity of Causes:** Business phenomena are never driven by a single factor. For instance, company sales are simultaneously influenced by pricing, competitor actions, consumer sentiment, advertising, and macroeconomics.
3. **Numerically Expressed:** Qualitative descriptions such as "rich", "poor", "high quality", or "efficient" cannot be statistically analyzed until quantified via rating scales or measurable indicators.
4. **Enumerated or Estimated with Reasonable Accuracy:** Data should either be accurately counted (for manageable populations) or estimated using robust sampling theory. Absolute mathematical accuracy is rarely attainable; a reasonable, standard margin of error is accepted.
5. **Collected in a Systematic Manner:** Haphazard data collection introduces severe biases. Reliable data requires structured sampling frames, validated questionnaires, and controlled experiments.
6. **Pre-determined Purpose:** Data must be compiled with a clearly formulated objective; otherwise, effort and organizational resources are wasted on irrelevant metrics.
7. **Placed in Relation to Each Other:** Data points must be comparable with respect to time, geography, or categorical units (homogeneity of data) to allow meaningful comparative inferences.

Q.3 Explain positive, negative, and symmetrical distributions with suitable diagrams and examples.

[CO1 | PEO2 | PO3 | PSO2 — 5.5 Marks]

The shape of a frequency distribution describes how data points are dispersed around

the central value. Based on the symmetry or direction of the elongated tail, distributions are classified into three types:



1. Symmetrical Distribution

- **Characteristics:** A bell-shaped, balanced curve where the left and right halves are exact mirror images of each other. The frequency rises smoothly to a peak and declines symmetrically. The values of mean, median, and mode are identical (**Mean = Median = Mode**), and skewness is zero (**Sk = 0**). The quartiles are equidistant from median: $Q_3 - M = M - Q_1$.
- **Managerial Example:** Standardized test scores (e.g., GMAT/CAT percentiles) or packaging weights in an automated bottling facility operating under stable statistical control.

2. Positively Skewed Distribution (Right-Skewed)

- **Characteristics:** The distribution curve stretches with a long, thin tail extending towards the right (higher positive values). The bulk of observations cluster at the lower end. Because extreme high values pull the mean upward, the mathematical order is **Mean** > **Median** > **Mode**, and skewness is positive ($Sk > 0$). Furthermore, $(Q_3 - M) > (M - Q_1)$.
- **Managerial Example:** Household income or corporate executive compensation. A vast majority earn average salaries, while a small elite earn exceptionally high figures, skewing the distribution to the right.

3. Negatively Skewed Distribution (Left-Skewed)

- **Characteristics:** The tail of the distribution extends towards the left (lower/negative values), while the mass of the data is concentrated on the right. Extreme low values pull the arithmetic mean downward, yielding the relationship **Mean** < **Median** < **Mode**, and skewness is negative ($Sk < 0$). Furthermore, $(Q_3 - M) < (M - Q_1)$.
- **Managerial Example:** Human lifespan in developed nations or scores on an excessively easy corporate compliance test where most employees score near 100% and only a few score low.

Q.4 From the data given below, find the Mean, Median, and Mode:

[CO1 | PEO1 | PO3 | PSO2 — 5.5 Marks]

Given Data:Central size (Mid-values, m): 35, 45, 55, 65, 75, 85Frequencies (f): 18, 37, 45, 27, 15, 8

Data Formulation: The difference between consecutive central values is $h = 45 - 35 = 10$. Hence, the class intervals are formed by taking $m \pm h/2 = m \pm 5$:

30–40, 40–50, 50–60, 60–70, 70–80, and 80–90.

Calculation Table:

Class Interval	Mid-value (m)	Frequency (f)	$d' = (m - 55)/10$	$f \cdot d'$	$f \cdot m$	Cumulative Freq. (cf)
30 – 40	35	18	–2	–36	630	18
40 – 50	45	37	–1	–37	1,665	55
50 – 60	55 (A)	45	0	0	2,475	100
60 – 70	65	27	+1	+27	1,755	127
70 – 80	75	15	+2	+30	1,125	142
80 – 90	85	8	+3	+24	680	150
Total	—	N = 150	—	$\sum f \cdot d' = +8$	$\sum f \cdot m = 8,330$	—

Step 1: Calculation of Arithmetic Mean ($\bar{X}$)Using Assumed Mean $A = 55$ and class interval width $h = 10$:

$$\bar{X} = A + [(\sum f \cdot d') / N] \times h = 55 + [8 / 150] \times 10 = 55 + 80 / 150 = 55 + 0.5333 = 55.5333$$

Verification via Direct Method: $\bar{X} = \sum f \cdot m / N = 8,330 / 150 = 55.5333$.**Step 2: Calculation of Median (M)**Position of Median = $N / 2 = 150 / 2 = 75$ -th observation.Looking at the cumulative frequencies, cf just greater than or equal to 75 is **100**, which corresponds to the class interval **50 – 60**.Therefore, the **Median Class = 50 – 60**.Here: Lower limit (L) = 50, Preceding cumulative frequency (cf_0) = 55, Frequency of median class (f) = 45, Width (h) = 10.

$$\text{Median} = L + [(N/2 - cf_0) / f] \times h = 50 + [(75 - 55) / 45] \times 10 = 50 + [20 / 45] \times 10 = 50 + 4.4444 = 54.4444$$

Step 3: Calculation of Mode (Z)The maximum frequency is $f = 45$, which lies in the class interval **50 – 60**.Therefore, the **Modal Class = 50 – 60**.

Here: Lower limit (L) = 50, Modal frequency (f_1) = 45, Preceding frequency (f_0) = 37, Succeeding frequency (f_2) = 27, Width (h) = 10.

$$\text{Mode} = L + [(f_1 - f_0) / (2 \cdot f_1 - f_0 - f_2)] \times h$$

$$\text{Mode} = 50 + [(45 - 37) / (2(45) - 37 - 27)] \times 10 = 50 + [8 / (90 - 64)] \times 10 = 50 + [8 / 26] \times 10 = 50 + 3.0769 = 53.0769$$

Final Results for Q.4:

- Mean (X) = 55.53 (or 55.5333)
- Median (M) = 54.44 (or 54.4444)
- Mode (Z) = 53.08 (or 53.0769)

Q.5 Find the Mean, Mode, Standard Deviation, and Pearson's Coefficient of Skewness for the following data: [CO1 | PEO1 | PO3 | PSO2 — 5.5 Marks]

Given Data:

Year under: 10, 20, 30, 40, 50, 60

No. of persons: 15, 32, 51, 78, 97, 100

Data Formulation: The table presents cumulative "less than" data. The true continuous frequency distribution is obtained by successive subtractions:

- 0 – 10: 15
- 10 – 20: 32 – 15 = 17
- 20 – 30: 51 – 32 = 19
- 30 – 40: 78 – 51 = 27
- 40 – 50: 97 – 78 = 19
- 50 – 60: 100 – 97 = 3

Calculation Table:

Age Group (Years)	Mid-value (m)	Frequency (f)	$d' = (m - 25)/10$	$f \cdot d'$	$f \cdot (d')^2$	$f \cdot m$	cf
0 – 10	5	15	-2	-30	60	75	15
10 – 20	15	17	-1	-17	17	255	32
20 – 30	25 (A)	19	0	0	0	475	51
30 – 40	35	27	+1	+27	27	945	78
40 – 50	45	19	+2	+38	76	855	97
50 – 60	55	3	+3	+9	27	165	100
Total	—	N = 100	—	$\sum f \cdot d' = +27$	$\sum f \cdot (d')^2 = 207$	$\sum f \cdot m = 2,770$	—

Step 1: Arithmetic Mean (X)

Using Assumed Mean $A = 25$ and width $h = 10$:

$$X = A + [(\sum f \cdot d') / N] \times h = 25 + [27 / 100] \times 10 = 25 + 2.7 = 27.70 \text{ years}$$

Step 2: Mode (Z)

The highest frequency is $f = 27$, occurring in the class interval **30 – 40**.

Modal Class = **30 – 40**.

Lower limit (L) = 30, Modal frequency (f_1) = 27, Preceding frequency (f_0) = 19, Succeeding frequency (f_2) = 19, Width (h) = 10.

$$\text{Mode} = L + [(f_1 - f_0) / (2 \cdot f_1 - f_0 - f_2)] \times h$$

$$\text{Mode} = 30 + [(27 - 19) / (2(27) - 19 - 19)] \times 10 = 30 + [8 / (54 - 38)] \times 10 = 30 + [8 / 16] \times 10 = 30 + 5.0 = 35.00 \text{ years}$$

Step 3: Standard Deviation (σ)

Using the step-deviation formula:

$$\sigma = h \times \sqrt{[(\sum f(d')^2 / N) - (\sum f \cdot d' / N)^2]}$$

$$\sigma = 10 \times \sqrt{[(207 / 100) - (27 / 100)^2]} = 10 \times \sqrt{[2.07 - (0.27)^2]} = 10 \times \sqrt{[2.07 - 0.0729]}$$

$$\sigma = 10 \times \sqrt{[1.9971]} = 10 \times 1.413188 = 14.1319 \text{ years}$$

Variance: $\sigma^2 = 199.71$.

Step 4: Pearson's Coefficient of Skewness (Sk_p)

Primary Formula (using Mode):

$$Sk_p = (Mean - Mode) / \sigma = (27.70 - 35.00) / 14.1319 = -7.30 / 14.1319 = -0.5166$$

Alternative Formula (using Median):

Median Class: $N/2 = 50$ lies in class 20 – 30 ($cf = 51$).

Median = $20 + [(50 - 32) / 19] \times 10 = 20 + [18 / 19] \times 10 = 20 + 9.4737 = 29.4737$.

$$Sk_p = [3 \cdot (Mean - Median)] / \sigma = [3 \cdot (27.70 - 29.4737)] / 14.1319 = [3 \cdot (-1.7737)] / 14.1319 = -0.3765$$

Interpretation: The negative coefficient of skewness confirms that the distribution is **moderately negatively skewed (skewed to the left)**, with a longer tail towards younger ages.

Final Results for Q.5:

- Arithmetic Mean ($\bar{X}$) = 27.70 years
- Mode (Z) = 35.00 years
- Standard Deviation (σ) = 14.13 years (or 14.1319)
- Pearson's Coefficient of Skewness (Sk_p) = -0.517 (using Mode) / -0.377 (using Median)